

Thibaut Montes

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PROFESSIONAL EXPERIENCE

QUANTITATIVE RESEARCHER AND DEVELOPER | ABU DHABI INVESTMENT AUTHORITY (ADIA)

July 2023 - Now | Abu Dhabi, United Arab Emirates

ADIA is one of the largest sovereign wealth funds in the world. I work in the QRD team whose objective is to develop and implement investment strategies while advancing work to unify its data platform and enhance its engineering infrastructure. My main tasks are to:

- ◇ Conduct research and deploy advanced pipelines for processing textual data in systematic investment strategies.
- ◇ Leverage **NLP** and **LLMs** to enhance model training and feature extraction processes.
- ◇ Apply data-driven insights to support Market Neutral and Directional strategies, both systematic and semi-systematic approaches.

SENIOR MACHINE LEARNING RESEARCHER | TALKWALKER

Mai 2021 - June 2023 | Luxembourg, Luxembourg

Talkwalker is a Consumer Intelligence Acceleration platform for brands to drive business impact and revenue. I work in the **AI & Machine Learning** team responsible to extract information from raw data coming from social medias (text, images, podcasts, videos, etc). More specifically, I use deep neural networks in order to solve **NLP** related problems. Here are some examples of my projects:

- ◇ Train models using **pre-trained multilingual transformers models** for aspect-based sentiment analysis or emotion prediction.
- ◇ **Meta and Active Learning** based training to improve classifiers accuracy when fewer labelled documents are available.

DESK STRAT | DEUTSCHE BANK

Sep 2020 - April 2021 | London, United-Kingdom

I worked on the Synthetic Desk's (ETFs and Indices) migration to Kannon (New P&L and Risk tool used by trading). My main tasks was to improve PV and Risk computations of ETFs and indices.

RESEARCH ADVISOR | THE INDEPENDENT CALCULATION AGENT (NOW OPENSEE)

May 2016 - Mar 2020 | Paris, France

First as an intern, then as a PhD candidate, I worked in collaboration with ICA on the following projects:

- ◇ Optimization of the analytic library using Optimal Quantization (Pricing of Exotic Options in the interest rate world).
- ◇ Identifying and *killing* bias in xVA computation using Multilevel Monte-Carlo methods.

EDUCATION

PHD IN NUMERICAL PROBABILITY | LPSM | SORBONNE UNIVERSITY

Mar 2017 - Jun 2020

I was under the direction of **Gilles Pagès** and **Vincent Lemaire** at the LPSM and the supervision of **Jean-Michel Fayolle** at ICA.

During my PhD, I made contributions to the theoretical study and financial applications of **Optimal Quantization**, also known as K-means. I also had a keen interest for **Multilevel Monte-Carlo** methods and **Stochastic Algorithms**.

- ◇ First, I focused on the numerical optimization of the problem (fixed point search and gradient descent) in order to efficiently build such quantizers.
- ◇ Then, I applied this numerical method for the pricing of PRDC bermudan options or the study of a Stationary Heston model.

RESEARCH MASTER IN PROBABILITY AND FINANCE (MASTER EL-KAROUI - WITH HONORS) |

SORBONNE UNIVERSITY

Sep 2014 - Jun 2016

BACHELOR DEGREE IN MATHEMATICS (WITH HONORS) | AIX-MARSEILLE UNIVERSITY

Sep 2011 - Jun 2014

SKILLS

PROGRAMMING

- ◇ **Python** (Pandas, Huggingface, PyTorch, ONNX, NumPy, pytest, Pybind11, ...)
- ◇ **SQL** (Redshift)
- ◇ **C++** (Creation of libraries during my PhD)
- ◇ Tools used often: **Git** ◇ **Airflow** ◇ **S3**
- ◇ Basics knowledge: **Java** ◇ **Scala** ◇ **Kafka** ◇ **MongoDB** ◇ **Docker**

LANGUAGES

French: native – **English:** fluent

PUBLICATIONS

- ◇ Lemaire, V., Montes, T. and Pagès G. (2022) **Stationary Heston model: Calibration and Pricing of exotics using Product Recursive Quantization.** *Quantitative Finance*, 2022.
- ◇ Fayolle, J.-M., Lemaire, V., Montes, T. and Pagès G. (2021) **Quantization-based Bermudan option pricing in the FX world.** *Journal of Computational Finance*, 2021.
- ◇ Lemaire, V., Montes, T. and Pagès G. (2020) **New weak error bounds and expansions for optimal quantization.** *Journal of Computational and Applied Mathematics*, 2020.